

Philippine Educational Attainment Analysis

A regional portfolio report using Philippine Statistics Authority data

This report compares the share of the population represented by college and post-baccalaureate categories across 17 matched Philippine regions. It is a descriptive snapshot, not a measure of change over time.

17 matched regions	22.23% regional mean	21.95 pp highest-lowest gap
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Headline finding

Regional attainment varies substantially. NCR records the highest value at 35.52%, while BARMM records the lowest at 13.57%, producing a gap of 21.95 percentage points.

Prepared as a clear, non-technical companion to the reproducible Google Colab notebook.

1. Purpose and approach

Why this analysis matters

National averages can hide meaningful differences in educational attainment. Comparing regions helps identify where attainment is relatively high or low and where deeper investigation may be useful.

Research questions

- Which regions have the highest educational attainment?
- Which regions have the lowest educational attainment?
- How does educational attainment vary across regions?
- Has educational attainment improved over time?
- Which regions improved the most?

Data and measure

The education data come from the Philippine Statistics Authority workbook *3_HGC and Literacy Statistical Table.xlsx*, worksheet *Table B*, stored in Google Drive. The notebook calculates the regional indicator as college plus post-baccalaureate counts divided by the extracted total, multiplied by 100.

Quality checks

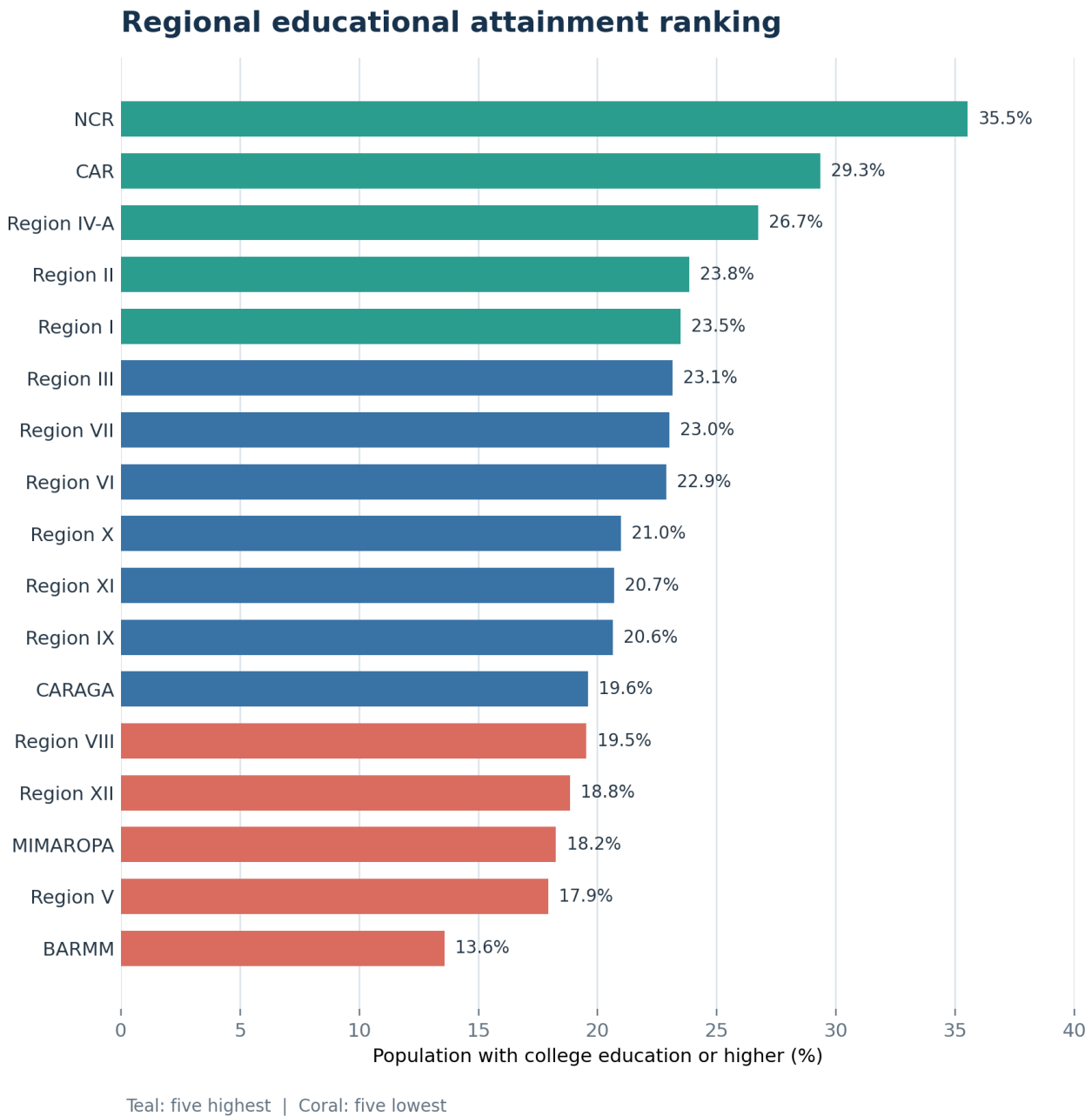
- Regional names were standardized before matching.
- The analysis checks missing values, duplicate rows, duplicate region labels, and whether percentages fall between 0 and 100.
- Source values were not modified or imputed.

Important reading note: The indicator should be interpreted within the population universe and category definitions of PSA Table B. The analysis shows regional differences but does not explain their causes.

2. Regional ranking

Purpose

This chart places all 17 matched regions on the same scale, making the highest, lowest, and middle values easy to compare.



Observation

NCR leads at 35.52%, followed by CAR at 29.34% and Region IV-A at 26.73%. BARMM has the lowest value at 13.57%.

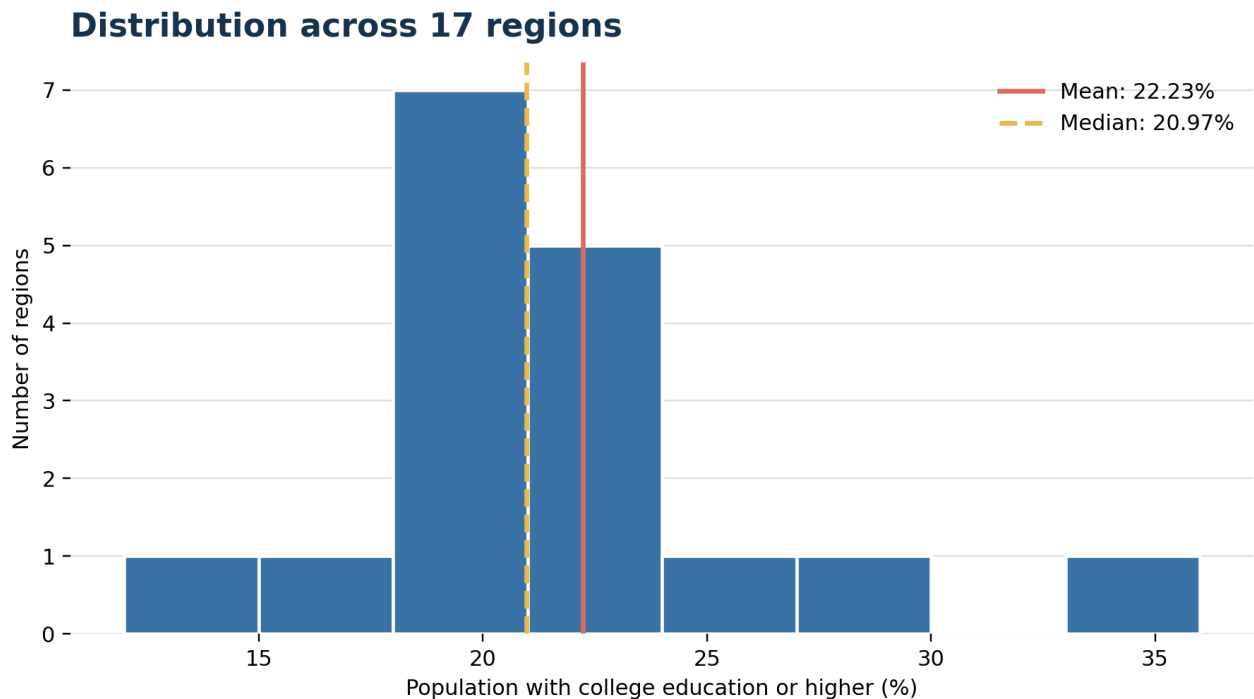
Interpretation

The ordering shows a wide regional spread rather than a uniform national pattern. Rankings are descriptive and should not be used as evidence of why one region differs from another.

3. Distribution of attainment

Purpose

The distribution chart shows where most regional values are concentrated and whether a small number of high values influence the overall average.



Observation

The mean is 22.23%, while the median is 20.97%. Most regions cluster around the high teens and low twenties, with several regions extending into the upper twenties and thirties.

Interpretation

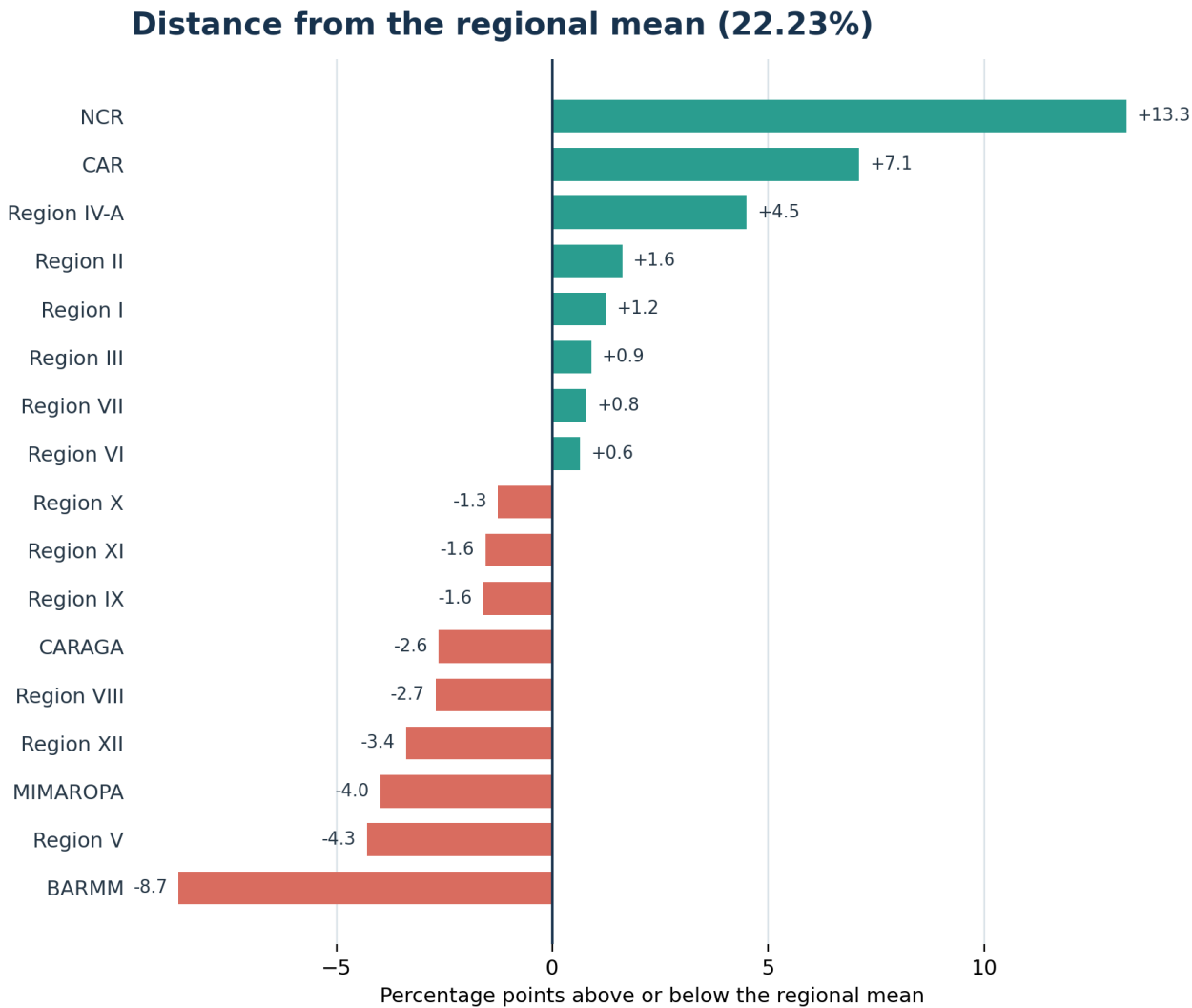
Because the mean is above the median, the small group of higher-attainment regions raises the overall average. A national average alone would therefore understate the size of regional differences.

At a glance: 8 regions are above the regional mean and 9 are below it. The full range is 21.95 percentage points.

4. Distance from the regional mean

Purpose

This view shows how far each region sits above or below the 22.23% regional mean. It highlights the scale of the difference without changing the original values.



Observation

NCR is about 13.3 percentage points above the mean. BARMM is about 8.7 points below it. CAR and Region IV-A also sit clearly above the center of the distribution.

Interpretation

The distances reinforce that the average is not representative of every region. Regional context is necessary when communicating the national education picture.

5. Highest and lowest regions

Highest educational attainment

Rank	Region	Attainment
1	NCR	35.52%
2	CAR	29.34%
3	Region IV-A	26.73%
4	Region II	23.85%
5	Region I	23.47%

NCR has the highest measured share. CAR and Region IV-A also stand above the regional mean. These results answer Research Question 1.

Lowest educational attainment

Rank	Region	Attainment
1	BARMM	13.57%
2	Region V	17.93%
3	MIMAROPA	18.25%
4	Region XII	18.84%
5	Region VIII	19.53%

BARMM has the lowest measured share, followed by Region V and MIMAROPA. These results answer Research Question 2 and identify where further contextual research may be especially valuable.

What the gap means

The 21.95-percentage-point difference between NCR and BARMM is large relative to the 22.23% regional mean. It shows substantial variation, answering Research Question 3. It does not show that region alone causes the difference.

6. Answers and limitations

Answers to the five research questions

- **Highest:** NCR, CAR, Region IV-A, Region II, and Region I.
- **Lowest:** BARMM, Region V, MIMAROPA, Region XII, and Region VIII.
- **Regional variation:** 13.57% to 35.52%, a range of 21.95 percentage points.
- **Improvement over time:** Cannot be answered from the available education snapshot.
- **Most improved regions:** Cannot be ranked without comparable education data for multiple years.

Why Questions 4 and 5 remain unanswered

The notebook contains only one comparable educational-attainment measurement. Its 2018, 2021, and 2023 fields come from a separate supplementary workbook and are not education measurements. Using them to claim educational improvement would mix different concepts and create an unsupported result.

Other limitations

- The calculation depends on hard-coded workbook rows and columns; a revised PSA layout may require updates.
- Regional boundaries and classifications must be harmonized before future year-to-year comparisons.
- The analysis is descriptive and does not control for demographics, migration, access to institutions, urbanization, or economic conditions.
- The exact PSA population universe and category definitions should be verified before external publication.

Analytical discipline matters: reporting that a question cannot yet be answered is stronger than substituting an unrelated measure.

7. Discussion and recommendations

What the findings suggest

The results establish a clear cross-sectional pattern: a small group of regions has relatively high college-level attainment, while several regions remain well below the regional mean. The pattern is useful for directing attention, but it is not a diagnosis of the underlying reasons.

Recommendations

- Use the lower-ranked regions as starting points for deeper, context-aware investigation rather than treating rank alone as a policy conclusion.
- Validate the Table B population universe, educational categories, and reference year before publication.
- Add comparable PSA education tables for at least one additional year before evaluating improvement.
- Pair attainment with relevant access, demographic, and socioeconomic indicators in future descriptive work.
- Harmonize changing regional boundaries before calculating trends.

Future work

A stronger follow-up study would calculate percentage-point changes from consistently defined education tables, compare trends by sex or age group where available, and include a complete data dictionary with official PSA download links and access dates.

Final takeaway

The available evidence shows meaningful regional differences in educational attainment. The clearest next step is not a more complex model, but better comparable education data across time.

References

Philippine Statistics Authority. *3_HGC and Literacy Statistical Table.xlsx*, Table B.

Philippine Statistics Authority. *Table 1a. CV_2018, 2021 and 2023p Average Annual Family Income, by Per Capita Income Decile Class and by Region, Province and HUCs.xlsx*, Table 1a. Retained in the notebook as supplementary exploratory material; not used for education trend conclusions.

Source workbooks are accessed through the project owner's Google Drive in Google Colab.